

# Star-autonomy and Mackey functors

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<sup>1</sup>Largely joint work with Brian Day and Elango Panchadcharam in 2006-7

## Two topics somewhat connected

1. Star autonomous categories as higher dimensional Frobenius monoids
2. Mackey functors as generalized group representations

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3. Robert Seely (1989) established a connection with Girard's linear logic. This ripened them for a diagrammatic calculus.
4. Brian Day and the speaker (2004) related  $*$ -autonomous categories to quantum groupoids and recognized that  $*$ -autonomous categories were higher-dimensional Frobenius algebras.

# Frobenius monoids

1. A  $k$ -algebra  $A$  is Frobenius when it is equipped with a linear function  $\sigma : A \otimes A \rightarrow k$  which acts as a counit for a duality  $A \dashv A$  in the monoidal category  $\mathbf{Vect}_k$  and satisfies  $\sigma(ab \otimes c) = \sigma(a \otimes bc)$ .

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2. The existence of any monoidal dual at all for  $A$  in  $\mathbf{Vect}_k$  implies  $A$  is finite dimensional and that the so-called dual space  $A^* = \mathbf{Vect}_k(A, k)$  is a monoidal dual. The formula satisfied by  $\sigma$  means it is determined by the linear function  $\varepsilon = \sigma(1 \otimes -) : A \rightarrow k$ . This leads to the following form of the definition.

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3. A monoid  $(A, I \xrightarrow{j} A, A \otimes A \xrightarrow{p} A)$  in a monoidal category  $\mathcal{V}$  is *Frobenius* when it admits a right dual  $A^*$  with unit  $I \xrightarrow{d} A \otimes A^*$  and it is equipped with a morphism  $A \xrightarrow{\varepsilon} I$  such that the composite

$$A \xrightarrow{1 \otimes d} A \otimes A \otimes A^* \xrightarrow{p \otimes 1} A \otimes A^* \xrightarrow{\varepsilon \otimes 1} A^*$$

is invertible.

# Frobenius monoidales

A monoidale (= pseudomonoid)

$$(A, I \xrightarrow{j} A, A \otimes A \xrightarrow{p} A, p(p \otimes 1) \xRightarrow{\alpha} p(1 \otimes p), p(j \otimes 1) \xRightarrow{\lambda} 1 \xleftarrow{\rho} p(1 \otimes j))$$

in a monoidal bicategory  $\mathcal{M}$  is *Frobenius* when it admits a right bidual  $A^*$  with unit  $I \xrightarrow{d} A \otimes A^*$  and it is equipped with a morphism  $A \xrightarrow{\varepsilon} I$  such that the composite

$$A \xrightarrow{1 \otimes d} A \otimes A \otimes A^* \xrightarrow{p \otimes 1} A \otimes A^* \xrightarrow{\varepsilon \otimes 1} A^*$$

is an equivalence.

# The bicategory $\mathcal{V}\text{-Mod}$

Let  $\mathcal{V}$  be a complete cocomplete symmetric closed monoidal category. Take the examples of  $\mathcal{V} = \text{Set}$  or  $\mathcal{V} = \text{Vect}_k$  for comfort. The objects of  $\mathcal{V}\text{-Mod}$  are  $\mathcal{V}$ -categories (ordinary categories and  $k$ -linear categories in the two examples).

The homcategories are defined by

$$\mathcal{V}\text{-Mod}(\mathcal{A}, \mathcal{B}) = [\mathcal{B}^{\text{op}} \otimes \mathcal{A}, \mathcal{V}]_0 ;$$

objects of these homs are called  $(\mathcal{V}\text{-})modules$ . Composition

$$\mathcal{V}\text{-Mod}(\mathcal{B}, \mathcal{C}) \times \mathcal{V}\text{-Mod}(\mathcal{A}, \mathcal{B}) \xrightarrow{\circ} \mathcal{V}\text{-Mod}(\mathcal{A}, \mathcal{C})$$

is defined by  $(\mathbf{N} \circ \mathbf{M})(C, A) = \int^B \mathbf{M}(B, A) \otimes \mathbf{N}(C, B)$ . The identity module  $\mathbf{1} : \mathcal{A} \rightarrow \mathcal{A}$  is the  $\mathcal{V}$ -valued hom functor of  $\mathcal{A}$ .

# The symmetric autonomous monoidal structure on $\mathcal{V}\text{-Mod}$

1. Tensor product  $\mathcal{A} \otimes \mathcal{B}$  is finite cartesian product of object sets and tensor product in  $\mathcal{V}$  of homs. The tensor unit  $\mathcal{I}$  has one object 0 and  $\mathcal{I}(0,0) = I$ .

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3. We identify each  $\mathcal{V}$ -functor  $F : \mathcal{A} \rightarrow \mathcal{B}$  with the module  $F_* : \mathcal{A} \rightarrow \mathcal{B}$  defined by  $F_*(B, A) = \mathcal{B}(B, FA)$ .  
Modules isomorphic to an  $F_*$  are said to **converge** to  $F$ .  
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4. The monoidal bicategorical dual of  $\mathcal{A}$  in  $\mathcal{V}\text{-Mod}$  is  $\mathcal{A}^{\text{op}}$ . The counit  $\mathcal{A} \otimes \mathcal{A}^{\text{op}} \xrightarrow{\mathbf{E}} \mathcal{I}$  is the switched hom  $\mathcal{V}$ -functor  $\mathcal{A} \otimes \mathcal{A}^{\text{op}} \xrightarrow{\mathbf{E}} \mathcal{V}$ .

# Cauchy completeness à la Lawvere

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6. Equivalence in  $\mathcal{V}\text{-Mod}$  is Morita equivalence. However, equivalence between Cauchy complete  $\mathcal{V}$ -categories in  $\mathcal{V}\text{-Mod}$  is equivalence in  $\mathcal{V}\text{-Cat}$  (since equivalences have right adjoints).

## Promonoidal $\mathcal{V}$ -categories

- ▶ Monoidales (= pseudomonoids) in  $\mathcal{V}\text{-Mod}^{\text{op}}$  are promonoidal  $\mathcal{V}$ -categories  $\mathcal{A}$  in the sense of Day (1970). The tensor product is a module  $\mathbf{P} : \mathcal{A} \rightarrow \mathcal{A} \otimes \mathcal{A}$  and the unit is a module  $\mathbf{J} : \mathcal{A} \rightarrow \mathcal{I}$ . As  $\mathcal{V}$ -functors,  $\mathbf{P} : \mathcal{A}^{\text{op}} \otimes \mathcal{A}^{\text{op}} \otimes \mathcal{A} \rightarrow \mathcal{V}$  and  $\mathbf{J} : \mathcal{A} \rightarrow \mathcal{V}$ .
- ▶ Suppose  $\mathcal{X}$  is a sufficiently cocomplete closed monoidal  $\mathcal{V}$ -category. The Day convolution monoidal structure on the  $\mathcal{V}$ -functor category  $[\mathcal{A}, \mathcal{X}]$  has unit  $\mathbf{J} \otimes I$  and tensor product

$$(F \star G)C = \int^{A,B} \mathbf{P}(A, B; C) \otimes FA \otimes GB .$$

- ▶ A *monoidal  $\mathcal{V}$ -category* is a promonoidal  $\mathcal{V}$ -category  $\mathcal{A}$  for which the modules  $\mathbf{J}^{\text{op}} : \mathcal{I} \rightarrow \mathcal{A}^{\text{op}}$  and  $\mathbf{P}^{\text{op}} : \mathcal{A}^{\text{op}} \otimes \mathcal{A}^{\text{op}} \rightarrow \mathcal{A}^{\text{op}}$  are convergent:

$$\mathbf{J}C \cong \mathcal{A}(I, C) , \quad \mathbf{P}(A, B; C) \cong \mathcal{A}(A \otimes B, C) .$$

## Frobenius promonoidal $\mathcal{V}$ -categories

- ▶ A *Frobenius* promonoidal  $\mathcal{V}$ -category is a Frobenius monoidal  $\mathcal{V}$ -category in  $\mathcal{V}\text{-Mod}^{\text{op}}$ .
- ▶ It consists of a promonoidal  $\mathcal{V}$ -category  $(\mathcal{A}, \mathbf{J}, \mathbf{P})$  together with a module  $\mathbf{N} : \mathcal{I} \rightarrow \mathcal{A}$  such that the composite module

$$\mathbf{S} : \mathcal{A}^{\text{op}} \xrightarrow{\mathbf{N} \otimes 1} \mathcal{A} \otimes \mathcal{A}^{\text{op}} \xrightarrow{\mathbf{P} \otimes 1} \mathcal{A} \otimes \mathcal{A} \otimes \mathcal{A}^{\text{op}} \xrightarrow{1 \otimes \mathbf{E}} \mathcal{A}$$

is an equivalence. Evaluating the composite, we obtain

$$\mathbf{S}(B, A) = \int^X \mathbf{N}X \otimes \mathbf{P}(A, B; X) .$$

$$\begin{aligned} \text{So } \int^B \mathbf{S}(B, A) \otimes \mathbf{J}B &\cong \int^{B, X} \mathbf{N}X \otimes \mathbf{P}(A, B; X) \otimes \mathbf{J}B \\ &\cong \int^X \mathbf{N}X \otimes \mathcal{A}(A, X) \\ &\cong \mathbf{N}A \end{aligned}$$

# Star-autonomous $\mathcal{V}$ -categories

- ▶ A *star-autonomous*  $\mathcal{V}$ -category is a monoidal  $\mathcal{V}$ -category which is Frobenius as a promonoidal  $\mathcal{V}$ -category and has **S** convergent:

$$\mathbf{S}(B, A) = \mathcal{A}(B, SA) .$$

- ▶ By our previous discussion, it is essentially without loss of generality to assume  $\mathcal{A}$  is Cauchy complete. Then the convergence of **S** to a  $\mathcal{V}$ -functorial equivalence  $S : \mathcal{A}^{\text{op}} \rightarrow \mathcal{A}$  is automatic.
- ▶ For  $\mathcal{A}$  star-autonomous,

$$\mathbf{N}A \cong \int^B \mathcal{A}(B, SA) \otimes \mathcal{A}(I, B) \cong \mathcal{A}(I, SA) .$$

- ▶ So the condition on the  $\mathcal{V}$ -functor  $S$  is:

$$\mathcal{A}(B, SA) (\cong \int^X \mathcal{A}(I, SX) \otimes \mathcal{A}(A \otimes B, X)) \cong \mathcal{A}(I, S(A \otimes B)) .$$

# Star-autonomy and closedness

## ► Proposition

*Star-autonomous  $\mathcal{V}$ -categories are closed monoidal.*

## Proof.

$\mathcal{A}(A \otimes B, C) \cong \mathcal{A}(SC, S(A \otimes B)) \cong \mathcal{A}(I, S(A \otimes B \otimes SC)) \cong$   
 $\mathcal{A}(B \otimes SC, SA) \cong \mathcal{A}(A, S^{-1}(B \otimes SC))$ , so  $\text{Lom}(B, C) = S^{-1}(B \otimes SC)$  is  
a left internal hom for  $\mathcal{A}$ . Exercise: obtain the right hom  $\text{Rom}(A, C)$ .  $\square$

## ► Proposition (M. Shulman, 2004)

*The unit from a symmetric monoidal closed category to the free symmetric star-autonomous category thereon is fully faithful.*

# Star-autonomy and convolution

## Proposition

Suppose  $\mathcal{A}$  is a small Frobenius promonoidal  $\mathcal{V}$ -category and  $\mathcal{X}$  is a cocomplete star-autonomous  $\mathcal{V}$ -category. Then the Day convolution monoidal  $\mathcal{V}$ -category  $[\mathcal{A}, \mathcal{X}]$  is star-autonomous with  $S(F) = S \circ F \circ S^{-1}$ .

## Proof.

For simplicity, we take the case where  $\mathcal{A}$  is star-autonomous. We will use  $\mathcal{A}(B \otimes C, S^{-1}I) \cong \mathcal{A}(I, S(B \otimes C)) \cong \mathcal{A}(C, SB)$  at the third step below. The calculation  $[\mathcal{A}, \mathcal{X}](\mathcal{A}(I, -) \otimes I, S(F * G)) \cong$   
 $\mathcal{X}(I, S((F * G)(S^{-1}I))) \cong \mathcal{X}(I, S \int^{B,C} \mathcal{A}(B \otimes C, S^{-1}I) \otimes FB \otimes GC)$   
 $\cong \mathcal{X}(I, S \int^{B,C} \mathcal{A}(C, SB) \otimes FB \otimes GC) \cong \int_B \mathcal{X}(I, S(FB \otimes GSB))$   
 $\cong \int_B \mathcal{X}(GSB, SFB) \cong \int_D \mathcal{X}(GD, SFS^{-1}D) \cong [\mathcal{A}, \mathcal{X}](G, S(F))$  gives the result since  $S$  is clearly an equivalence. □

# Autonomous implies star-autonomous

- ▶ Recall that a monoidal  $\mathcal{V}$ -category  $\mathcal{A}$  is called *autonomous*<sup>2</sup> when each object  $A$  has a right dual  $A^\vee$  and a left dual  $A^\wedge$ . We have adjunctions

$$A^\wedge \otimes - \dashv A \otimes - \dashv A^\vee \otimes -$$

of endo- $\mathcal{V}$ -functors on  $\mathcal{A}$ .

- ▶ Right dual becomes a  $\mathcal{V}$ -functor  $(-)^\vee : \mathcal{A}^{\text{op}} \rightarrow \mathcal{A}$  with left dual  $(-)^\wedge$  as inverse equivalence.
- ▶ It is easy to see that  $(A \otimes B)^\vee \cong B^\vee \otimes A^\vee$  and  $I^\vee \cong I$ .
- ▶ We have  $\mathcal{A}(I, (A \otimes B)^\vee) \cong \mathcal{A}(I, B^\vee \otimes A^\vee) \cong \mathcal{A}(B, A^\vee)$ .
- ▶ So every autonomous  $\mathcal{V}$ -category is star-autonomous where  $S$  returns the right dual.

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<sup>2</sup>also “compact” and “rigid” are used however we are not assuming symmetry  $\equiv$

# Towards the original form of Mackey functors

- ▶ A Mackey functor for a group  $G$  can be thought of as a generalized representation of  $G$  or as a decategorified representation theory for  $G$ :

$$\text{Rep } G \subseteq \text{Mky } G \ni "K_0 \text{Rep } G".$$

- ▶ Let's try to make categorical sense of J.A. Green's original definition of a Mackey functor.
- ▶ For any cospan  $H \xrightarrow{i} G \xleftarrow{j} K$  in  $\text{Cat}$  and any cocomplete category  $\mathcal{X}$ , there is a 2-dim form of Chevalley-Beck which says

$$\text{Res}_j \circ \text{Lan}_i \cong \text{Lan}_q \circ \text{Res}_p$$

where  $H \xleftarrow{p} i/j \xrightarrow{q} K$  is the **comma category** span while the functor  $[G, \mathcal{X}] \xrightarrow{\text{Res}_j} [K, \mathcal{X}]$  is **restriction** along  $j$  and  $\text{Lan}_i$  is **left Kan extension** along  $i$ .

## Kan extensions involving groupoids

- ▶ Now suppose  $i$  and  $j$  are inclusions of subgroups  $H$  and  $K$  of  $G$ . Then  $i/j$  is a groupoid whose objects are the elements of  $G$ .
- ▶ Isomorphism classes of objects of  $i/j$  are in bijection with double cosets

$$KgH = \{kgh \mid k \in K, h \in H\}.$$

Let  $[K \backslash G / H] \subseteq G$  represent all double cosets  $KgH$  so that we have an equivalence of categories  $\sum_{g \in [K \backslash G / H]} (i \downarrow j)(g, g) \simeq i \downarrow j$ .

- ▶ With  $p_g, q_g$  the restrictions of  $p, q$ , the Kan extension formula gives

$$\text{Res}_j \circ \text{Lan}_i \cong \sum_{g \in [K \backslash G / H]} \text{Lan}_{q_g} \circ \text{Res}_{p_g}.$$

With  $K^g = g^{-1}Kg$  and  ${}^gH = gHg^{-1}$ ,  $p_g, q_g$  induce isomorphisms

$$(i/j)(g, g) \cong H \cap K^g \text{ and } (i/j)(g, g) \cong {}^gH \cap K.$$

# The Mackey Decomposition Theorem

Fix a field  $k$  and take  $\mathcal{X} = \text{Vect}_k$ . Representation theory notation is:

$$\text{Res}_H^G = \text{Res}_i \quad \text{and} \quad \text{Ind}_H^G = \text{Lan}_i .$$

Let  $\gamma_g$  denote the conjugation isomorphism  $H \cap K^g \cong {}^g H \cap K$ .

## Theorem (Mackey Decomposition)

$$\text{Res}_H^G \circ \text{Ind}_H^G \cong \sum_{g \in [K \backslash G / H]} \text{Ind}_{g H \cap K}^K \circ \text{Res}_{\gamma_g} \circ \text{Res}_{H \cap K^g}^H$$

Compare this with axiom 4 in the upcoming definition of Mackey functor.

# Green's definition

## Definition (J.A. Green, 1971)

For a group  $G$ , a **Mackey functor**  $M$  over  $k$  assigns

- ▶ to each  $H \leq G$ , a vector space  $M(H)$ ,
- ▶ to each  $K \leq H \leq G$ , a pair of linear functions  $t_K^H : M(K) \rightarrow M(H)$  and  $r_K^H : M(H) \rightarrow M(K)$ ,
- ▶ for  $H \leq G$  and  $g \in G$ , a linear isomorphism  $c_{g,H} : M(H) \xrightarrow{\cong} M({}^gH)$ ,

satisfying the following four axioms:

1.  $t_K^H t_L^K = t_L^H$  and  $r_L^K r_K^H = r_L^H$ ,
2.  $c_{g', {}^gH} c_{g,H} = c_{g'g,H}$  and  $c_{h,H} = 1_{M(H)}$ ,
3.  $c_{g,H} t_K^H = t_{gK}^{gH} c_{g,K}$  and  $c_{g,K} r_K^H = r_{gK}^{gH} c_{g,H}$
4.  $r_K^L t_H^L = \sum_{g \in [K \backslash L/H]} t_{gH \cap K}^K c_{g,H \cap K^g} r_{H \cap K^g}^H$ .

$t_K^H$  is called *transfer*, *trace* or *induction*,  $r_K^H$  is called *restriction*, and  $c_{g,H}$  is called a *conjugation map*.

# Frobenius Reciprocity and Green functors

With  $i : H \rightarrow G$  as before, suppose  $[H, \text{Vect}]$  and  $[G, \text{Vect}]$  are equipped with the pointwise tensor products. For  $V \in [G, \text{Vect}]$  and  $W \in [H, \text{Vect}]$ , there is an isomorphism

$$V \otimes \text{Lan}_i(W) \cong \text{Lan}_i(\text{Res}_i(V) \otimes W).$$

**Frobenius Reciprocity:**  $V \otimes \text{Ind}_H^G(W) \cong \text{Ind}_H^G(\text{Res}_H^G(V) \otimes W).$

This inspires the technical axiom 6 in the following.

## Definition

A **Green functor**  $A$  for  $G$  over  $k$  is a Mackey functor equipped with an algebra structure on each vector space  $A(H)$ ,  $H \leq G$ , satisfying

5.  $t_K^H, r_K^H, c_{g,K}$  are algebra morphisms,
6.  $a \cdot t_K^H(b) = t_K^H(r_K^H(a) \cdot b)$  and  $t_K^H(b) \cdot a = t_K^H(b \cdot r_K^H(a)).$

## Example

Each representation  $R \in \text{Rep}(H) = [G, \text{Vect}]$  of  $G$  determines a Mackey functor  $M_R$  with

$$M_R(H) = [G, \text{Set}](G/H, R).$$

## Example

There is a Green functor  $A$  for  $G$  with  $A(H)$  equal to the free vector space on the abelian group  $\text{K}_0\text{Rep}(H)$ .

## Example

There are examples on taking  $G$  to be a Galois group so that subgroups correspond to field extensions. This is fine for  $G$  finite. When  $G$  is infinite (profinite), the definition of Mackey functor was modified by Bley-Boltje (2001) with applications to number theory and algebraic geometry.

## Towards the second definition

- ▶ Green already recognized that Mackey functors consisted of a pair of functors, one contravariant, one covariant, defined on the category  $G\text{-Set}_{\text{conn}}$  of connected  $G$ -sets.
- ▶ The category  $G\text{-set}$  of finite  $G$ -sets for a finite  $G$  is the free finite coproduct completion  $\text{fam}(G\text{-Set}_{\text{conn}})$  of  $G\text{-Set}_{\text{conn}}$ . Dress (1973) gave the following definition for  $\mathcal{E} = G\text{-set}$ .

## Definition

Let  $\mathcal{C}$  be a small category with finite coproducts. A **Mackey functor**  $M : \mathcal{C} \rightarrow \mathbf{Vect}$  consists of two functors  $\mathcal{C} \xrightarrow{M_*} \mathbf{Vect}$  and  $\mathcal{C}^{\text{op}} \xrightarrow{M^*} \mathbf{Vect}$  which agree on objects and satisfy the following two axioms:

1. if the left below is a pullback then the right commutes;

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ g \downarrow & & \downarrow h \\ Z & \xrightarrow{k} & W \end{array}$$

$$\begin{array}{ccc} MX & \xrightarrow{M_* f} & MY \\ M^* g \uparrow & & \uparrow M^* h \\ MZ & \xrightarrow{M_* k} & MW \end{array}$$

2. if the left below is a coproduct then the right is a direct sum.

$$X \xrightarrow{i} Z \xleftarrow{j} Y$$

$$MX \xrightleftharpoons[M^* i]{M_* i} MZ \xrightleftharpoons[M_* j]{M^* j} MY$$

# Spans to the rescue!

- ▶ Henceforth  $\mathcal{V}$  will denote the closed monoidal category of commutative monoids.
- ▶  $\mathcal{E}$  will denote a lextensive category: it has finite coproducts and finite limits, and such that coproduct provides an equivalence  $\mathcal{E}/X \times \mathcal{E}/Y \simeq \mathcal{E}/X + Y$  for all  $X, Y \in \mathcal{E}$ .
- ▶ Let  $\text{Spn}\mathcal{E}$  denote the  $\mathcal{V}$ -category with the same objects as  $\mathcal{E}$  and with morphisms  $X \xrightarrow{[S]=[(s_1, S, s_2)]} Y$  the isomorphism classes of spans  $X \xleftarrow{s_1} S \xrightarrow{s_2} Y$  in  $\mathcal{E}$ ; composite  $X \xrightarrow{[S]} Y \xrightarrow{[T]} Z$  is induced by pullback  $X \xrightarrow{[S \times_Y T]} Z$ ; addition in the homs is induced by coproduct.
- ▶  $\text{Spn}\mathcal{E}(X + Y, Z) \cong \text{Spn}\mathcal{E}(X, Z) \times \text{Spn}\mathcal{E}(Y, Z)$  uses lextensivity while  $(\text{Spn}\mathcal{E})^{\text{op}} \cong \text{Spn}\mathcal{E}$  so we have a direct sum  $X \oplus Y = X + Y$ .

# The monoidal structure on $\mathrm{Spn}\mathcal{E}$ and its properties

- ▶ With product in  $\mathcal{E}$  as tensor product,  $\mathrm{Spn}\mathcal{E}$  is a symmetric monoidal  $\mathcal{V}$ -category.
- ▶ There are identity-on-objects functors  $\mathcal{E} \xrightarrow{(-)^*} \mathrm{Spn}\mathcal{E}$ ,  $\mathcal{E}^{\mathrm{op}} \xrightarrow{(-)^*} \mathrm{Spn}\mathcal{E}$  taking  $X \xrightarrow{f} Y$  to  $f_* = [(1_X, X, f)]$ ,  $f^* = [(f, X, 1_X)]$ .
- ▶ **Every object  $X$  of  $\mathrm{Spn}\mathcal{E}$  is canonically a Frobenius monoid:** every object  $X$  of the cartesian monoidal category  $\mathcal{E}$  is a comonoid which is taken to a monoid by  $(-)^*$ ; the unit  $d$  for  $X \dashv X$  is  $[1 \xleftarrow{!} X \xrightarrow{\delta} X \times X]$  while the morphism  $e$  in the definition of Frobenius is  $!_* : X \rightarrow 1$ .
- ▶  $\mathrm{Spn}\mathcal{E}(X \times Y, Z) \cong \mathrm{Spn}\mathcal{E}(Y, X \times Z)$ .
- ▶ In particular,  $\mathrm{Spn}\mathcal{E}$  is autonomous.

# Mackey functors as functors, Green functors as monoids

## Proposition (H. Lindner, 1976)

A Mackey functor on  $\mathcal{E}$  is a  $\mathcal{V}$ -functor  $M : \text{Spn}\mathcal{E} \rightarrow \text{Vect}$ ; the old  $M_*$  and  $M^*$  are just the composites of  $M$  with  $(-)_*$  and  $(-)^*$ .

- ▶ Let  $\text{Mky}\mathcal{E}$  denote the  $\mathcal{V}$ -functor  $\mathcal{V}$ -category  $[\text{Spn}\mathcal{E}, \text{Vect}]$ .
- ▶ It is symmetric closed monoidal by Day convolution:

$$(M * N)Z = \int^Y M(Z \times Y) \otimes NY, \quad [M, N]Z = \text{Mky}\mathcal{E}(M(Z \times -), N).$$

The unit  $J$  is the *Burnside functor* whose value at  $X$  is the free vector space on the commutative monoid under coproduct of isomorphism classes of objects of  $\mathcal{E}/X$ .

## Proposition

*Green functors are monoids in  $\text{Mky}\mathcal{E}$ .*

# Finite Mackey functors

Let  $\text{Mky}_{\text{fin}} \mathcal{E}$  denote the full sub- $\mathcal{V}$ -category of  $\text{Mky} \mathcal{E}$  consisting of those  $M$  for which each  $MX$  is finite dimensional.

## Theorem

*Suppose  $\mathcal{E}$  is a lexextensive category for which every object is a coproduct of objects from a given finite set  $\mathcal{C}$  of objects. Then  $\text{Mky}_{\text{fin}} \mathcal{E}$  is closed under the symmetric closed monoidal structure on  $\text{Mky} \mathcal{E}$ . Moreover,  $\text{Mky}_{\text{fin}} \mathcal{E}$  is star-autonomous with  $S(M)X = (MX)^*$ .*

The point is that the coproducts mentioned become direct sums in  $\text{Spn} \mathcal{E}$  (and so are preserved by Mackey functors) and the homs in  $\text{Spn} \mathcal{E}$  become finitely generated commutative monoids. This means that the coends and ends involved in the convolution structure can be calculated over a finite category.

# The monoidal centre of $\mathbf{Mky}\mathcal{E}$

- ▶ For a finite group  $G$ , the category  $\mathcal{E} = [G, \mathbf{Set}_{\text{fin}}]$  of finite  $G$ -sets satisfies the hypothesis of the Theorem by taking  $\mathcal{C}$  to consist of the connected  $G$ -sets  $G/H$  for  $H \leq G$ .
- ▶ We proved that the monoidal centre  $\mathcal{Z}\mathbf{Mky}\mathcal{E}$  was a functor  $\mathcal{V}$ -category again. However, Tambara went further.

## Theorem (D. Tambara, 2008)

*With  $G$  a finite group and  $\mathcal{E} = [G, \mathbf{Set}_{\text{fin}}]$ , the centre  $\mathcal{Z}\mathbf{Mky}\mathcal{E}$  is equivalent to  $\mathbf{Mky}\mathcal{F}$  where  $\mathcal{F}$  is the finite coproduct completion of the arrow category of the full subcategory  $\mathcal{C}$  of connected objects of  $\mathcal{E}$ .*

# Mackey functors as generalized representations

- ▶ For a finite group  $G$ , put  $\text{Mky}_{\text{fin}}(G) = \text{Mky}_{\text{fin}}[G, \text{Set}_{\text{fin}}]$  and  $\text{Rep}_{\text{fin}}(G) = [G, \text{Vect}_{\text{fin}}]$ .
- ▶ Let  $k : [G, \text{Set}_{\text{fin}}] \rightarrow \text{Rep}_{\text{fin}}(G)$  take a  $G$ -set  $X$  to the vector space with  $X$  as basis and the linearized action.
- ▶ This extends to a strong monoidal  $\mathcal{V}$ -functor

$$k_* : (\text{Spn}[G, \text{Set}_{\text{fin}}])^{\text{op}} \rightarrow \text{Rep}_{\text{fin}}(G)$$
$$k_*(X \xleftarrow{u} S \xrightarrow{v} Y) : kY \rightarrow kX, \quad y \mapsto \sum_{v(s)=y} u(s).$$

- ▶ Applying the usual Kan business, we obtain a fully faithful, right adjoint, strong monoidal, closed under exponentiation,  $\mathcal{V}$ -functor

$$\text{Rep}_{\text{fin}}(G) \rightarrow \text{Mky}_{\text{fin}}(G).$$

# Thank you!

$$M \xrightarrow[\Delta]{\text{CoACT}} MQ$$